

Sarah Thomas

+91-9993137915 | sarahthomas29@gmail.com | linkedin.com/in/sarah-thomas-40a301289 | github.com/THOMAS-SARAH

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science – Specialization in Cloud Computing and Automation

September 2023 – May 2027

CGPA: 8.81/10

St. Joseph's Convent Sr. Sec. School, Sagar, Madhya Pradesh

All India Senior School Certificate Examination

March 2022 – May 2023

CGPA: 7.88/10

St. Joseph's Convent Sr. Sec. School, Sagar, Madhya Pradesh

All India Secondary School Examination

March 2020 – June 2021

CGPA: 9.2/10

TECHNICAL SKILLS

Cloud & DevOps: AWS (Lambda, S3, API Gateway, EC2, IAM, CloudWatch), Docker, Docker Compose, Prometheus (Metrics Telemetry), Terraform, CI/CD (via GitHub Actions), Git, GitHub, Boto3 (AWS SDK)

Languages & Frameworks: Java, Python, Linux/Bash, FastAPI, REST APIs, AIOps, Scikit-Learn, OpenCV

Databases & Tools: MongoDB, SQL Server, OpenAI API, Pandas, Matplotlib

PROJECTS

AI Incident & Log Analysis Assistant

Python, FastAPI, OpenAI API

github.com/THOMAS-SARAH/exception_analyzer

- Developed a RESTful AI incident response microservice using FastAPI and OpenAI to process server log traces and generate automated Root-Cause Analysis (RCA) reports.
- Implemented structured prompt engineering workflows to analyze system telemetry errors and output actionable remediation steps for operational teams.

Hybrid Ensemble Model for Enhanced Cancer Cell Classification

Python, Scikit-Learn, OpenCV, Pandas, Matplotlib

github.com/THOMAS-SARAH/CANCER_CELL_DETECTION

- Implemented a hybrid ensemble learning model combining multiple classifiers to distinguish cancerous from non-cancerous cells in medical imaging data.
- Optimized image preprocessing and feature extraction pipelines using OpenCV, improving model input quality and boosting overall prediction reliability.
- Evaluated model performance using accuracy, AUC, and standard classification metrics, visualizing comparative results with Pandas and Matplotlib to support model selection and validation.

AI-Driven Cloud Resource & Cost Optimizer

Prometheus, Node Exporter, Terraform, AWS, Linux/Bash, FastAPI github.com/THOMAS-SARAH/multi-region-aws-telemetry-pipeline

- Engineered a real-time infrastructure telemetry pipeline using Prometheus and Node Exporter to monitor system metrics, CPU/memory utilization, and workload health across 2+ simulated cloud instances while testing.
- Integrated Prometheus scraping endpoints with a FastAPI backend to ingest live telemetry data for automated rule-based resource optimization and anomaly detection.
- Provisioned multi-region cloud infrastructure using modular Terraform templates and built a traffic and failure simulation framework to stress-test system monitoring workflows.

CERTIFICATIONS AND EXTRA-CURRICULAR

Certifications

- ServiceNow University – Virtual Internship Program
- NPTEL – Internet Of Things, Cloud Computing and Distributed Systems
- AWS Educate – Introduction to Generative AI

Extra Curriculars

- Education Minister in School Council
- Best Girl Student of my Batch 2022-23
- Club Coordinator of E-Green Club